

SIDDHARTH KALYANI

AI/GenAI & Forward-Deployed Engineer | LLM Systems & Backend Engineering

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PROFESSIONAL SUMMARY

AI/GenAI & Forward-Deployed Engineer with 2+ years of experience architecting and shipping production LLM systems on Google Vertex AI and Gemini. Specializes in multi-agent RAG systems, GraphRAG-based retrieval, and production caching strategies, leading end-to-end delivery of 10+ AI solutions and 5+ production AI pipelines from stakeholder requirements and system architecture through deployment and client demos. Cut LLM inference costs by up to 80% and manual document-review effort by up to 95%, while processing 1B+ LLM tokens and 1M+ pages of unstructured documents (RFPs, healthcare literature) in production.

TECHNICAL SKILLS

Programming Languages: Python (async/await), SQL, JavaScript

Generative AI & LLMs: Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), GraphRAG, Knowledge Graphs, Multi-Agent Systems, Agentic AI, Model Routing, Structured Output, LLM Evaluation, Semantic Search, Embeddings

GenAI Platforms & Tools: Google Vertex AI, Gemini (2.5–3.6 family), Vertex AI Batch Prediction API, Vertex AI Context Caching API, Ollama (local LLM deployment), Qwen 3.6 35B, LangChain, Google Agent Builder, LLMOps

Backend Engineering & Frameworks: FastAPI, REST APIs, System Architecture, LlamaIndex, SQLAlchemy (async), Alembic, Concurrency & Fault-Tolerant Design (Retry/Backoff), Pipeline Orchestration

Databases & Vector Stores: PostgreSQL, pgvector

Cloud & Infrastructure: Google Cloud Platform (Cloud Storage), AWS (Secrets Manager), Linux (production server administration)

Data & Automation: Pandas, Playwright, SeleniumBase, Document Processing (PDF/DOCX), Git

PROFESSIONAL EXPERIENCE

Argusoft India Ltd, Gandhinagar, Gujarat 382016

Jul 2024 - Present

AI/GenAI & Forward-Deployed Engineer

- Bidstream (RFP Discovery, Validation & Proposal Automation Platform) – Multi-Agent Proposal Generation:** Architected an autonomous multi-agent system (Google ADK, Vertex AI) with Planner, Writer, and Reviewer agents using RAG, multimodal ingestion, and context caching — cutting manual proposal rework from ~50% of sections to ~5% minor edits.
- Proposal Scoring Engine:** Built a 6-stage agentic AI pipeline for rubric extraction, compliance auditing, and persona-driven evaluation, with async orchestration and prompt caching — reducing manual compliance review effort by 95%.
- Batched AI Processing & Prompt Optimization:** Developed a batched validation pipeline (Vertex AI Batch API, GCS, SHA-256 fingerprinting) that cut validation costs 50%, plus 30% via dedup; tuned prompts on 20K labeled rows to reach 95%+ accuracy.
- LLM Caching Infrastructure:** Designed a production-grade LLM caching platform on Vertex AI's CachedContent API, cutting generation costs 80%; standardized 100+ prompt templates for a further 20% reduction via implicit caching.
- LLMOps & Observability:** Implemented an observability layer tracking 1B+ cached and uncached tokens, quality metrics, and costs across 5+ production AI pipelines, with daily PostgreSQL reports guiding cost/quality tuning.
- Large-Context & Multimodal Document Processing:** Engineered a fault-tolerant chunking pipeline and native multimodal ingestion (PDF/DOCX to Gemini) for 1M+ token documents; optimized document scoping and media resolution to cut latency 60% and token costs 50% while improving accuracy — replacing an earlier LlamaIndex/pgvector RAG approach.
- GraphRAG Search Layer:** Delivered a GraphRAG-based search system enabling natural-language structured queries across 100,000+ RFP insights, replacing noisy vector retrieval to cut irrelevant LLM context.
- Data Collection Platform:** Scaled 30+ Playwright/SeleniumBase scrapers across government and enterprise portals, adding Gemini-based multimodal CAPTCHA-solving for challenge-response sites.

- **Production Infrastructure:** Managed demo (Windows) and production (Linux, DB, CI/CD) environments with an average deployment cadence of 2x/week, ensuring reliable uptime for client-facing systems.
- **Medplat (Healthcare RAG Platform) – Clinical Question-Answering:** Built an advanced RAG pipeline over medical reference literature to power accurate, source-grounded clinical question-answering.
- **Local LLM Deployment:** Deployed Qwen 3.6 35B locally via Ollama on the company’s RTX 6000 GPU to keep PHI processing fully on-premises.
- **HIPAA Compliance:** Designed HIPAA-compliant LLM processing workflows for handling protected health information, incorporating appropriate data-handling and access-control safeguards.
- **MeghDoot (Internal Messaging Platform) – Audio Signal Pipeline:** Engineered an audio-processing pipeline (noise reduction, silence trimming, speech-to-text transcription, and speaker diarization) to auto-generate searchable conversation transcripts for an internal Slack/Teams-style messaging platform.

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Jan 2024 - Jun 2024

AI/GenAI Engineer

- Completed an intensive 2-month AI/ML upskilling program before joining the RFP automation team.
- Built Playwright-based scrapers for RFP data collection, then engineered a SeleniumBase-based replacement pipeline after Playwright proved unable to reliably handle advanced bot-detection systems (e.g., Cloudflare) on key procurement portals, restoring reliable data collection.
- Contributed to the team’s earliest AI-pipeline development, laying groundwork later extended into the production systems above.
- Built HealthSync, a diet- and health-tracking web app (Angular 15, Node.js, PostgreSQL) as a proof-of-concept covering calorie/BMI tracking and verified-diet workflows.

EDUCATION

Vishwakarma Government Engineering College <i>B.Tech, Information Technology / CGPA: 9.30/10</i> Chandkheda, Ahmedabad, Gujarat 382424	2021 - 2024
Tolani FG Polytechnic <i>Diploma, Computer Engineering / CPI: 9.94/10</i> Gandhidham, Kutch, Gujarat 370205	2018 - 2021
St. Xavier’s High School <i>SSC / 84.53%</i> Gandhidham, Gujarat 370205	2017 - 2018

AWARDS & ACHIEVEMENTS

- **1st Place, Argusoft Internal 8-Hour AI Hackathon:** built a conversational form-filling chatbot with real-time input validation, mid-dialogue slot correction, and red-team-hardened scoring prompts to block buzzword-based gaming.
- Top 5 Finish, Machine Learning Competition at Argusoft
- 2nd Prize, TechXIT 2022 Web Developer Competition at VGEC

CERTIFICATIONS

- Meta Front-End Developer, Specialization in React (Coursera)